

Part B.

Question No 3.

• Problem Analysis Chart (PAC).

Input	Processing	Output	Decision
• N (Total students)	• loop N lines (outer loop)	• sum (Total Mark)	• Subject failure Any mark < 33
• S subjects marks per student (mark1, mark2, mark3, mark4, mark5)	• loop S lines per student (inner loop) • get marks	• average (Average marks)	⇒ has deficiency = True
	• Sum the S marks • calculate avg = sum/S	• classification (Final result)	• check if has deficiency == TRUE ⇒ "Fail - Subject Deficiency"
	• check if entered mark < 33		• Other wise check average ≥ 80 ⇒ "Distinction"
	• Determine classification based on avg		average ≥ 60 ⇒ "Pass" average < 60 ⇒ "Fail"

2. IPO Chart.

Input	Processing	Output
• N	1) Prompt user for total student N.	• average
• mark(s) (marks per student)	1) for each student 1 to N 2) Set sum = 0, has deficiency = FALSE 3) READ marks a) sum = sum + marks b) if mark < 33 then has deficiency = TRUE 4) average = sum / N 5) if has deficiency == TRUE then classification = "Distinction" classification = "Fail! Subject Deficiency" ELSE if average >= 80 THEN classification = "Distinction" ELSE if average >= 60 THEN classification = "PASS" ELSE classification = "FAIL" END IF 6) Output average and classifications.	• classification

3- Algorithm:

1) Start

2) Input Total number of student, N .

3) Set student counter $i = 1$.

4) Repeat while $i \leq N$

- Set $sum = 0$

- Set has Deficiency = FALSE

- Set Subject counter $j = 1$

- Repeat while $j \leq 5$

- Input marks

- $sum = sum + mark$

- if $mark < 33$

- set has Deficiency = TRUE

- Increment $j = j + 1$

- Calculate $average = sum / 5$

- if has Deficiency is TRUE

- Set classification = "Fail! - Subject Deficiency"

- ELSE if $average \geq 80$

- Set classification = "Distinction"

- ELSE if $average \geq 60$

- Set classification = "^{Pass}Distinction"

- ELSE

- Set classification = "FAIL!"

- Display average and classification for i

- Increment $i = i + 1$

5) END

4. Pseudo code

BEGIN

DECLARE N, i, j as integer
 DECLARE mark, sum, average as REAL
 DECLARE hasDeficiency
 DECLARE classification as STRING

PRINT "Enter total number of student"

READ N

FOR $i = 1$ TO N

DO

SET sum = 0

SET hasDeficiency = FALSE

PRINT "--- Student ---", i , "---"

FOR $j = 1$ TO S DO

PRINT "Enter marks for subject", j , ":"

READ mark

sum = sum + mark

If mark < 33 Then

hasDeficiency = TRUE

ENDIF

END FOR

average = sum / S

If hasDeficiency == TRUE THEN

classification = "Fail. Subject Deficiency"

ELSE IF (average >= 80) THEN

classification = "Distinction"

ELSE IF (average >= 60) THEN

classification = "Pass"

ELSE IF

classification = "Fail!"

END IF

PRINT "Average Mark", average

PRINT "Result", classification

END FOR

END

5. Flow chart:



